

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.datasets import cifar10

(cifar_x, _), _ = cifar10.load_data()

w = 1024
h = 768

new_w = int(w * 0.25)
new_h = int(h * 0.25)

print("Original Size:", w, "x", h)
print("Resized (25%):", new_w, "x", new_h)

resized_images = []
gray_images = []
blur_images = []
edge_images = []

kernel = tf.constant(
    [[[[1]], [[2]], [[1]]],
     [[2]], [[4]], [[2]]],
     [[1]], [[2]], [[1]]],
    dtype=tf.float32
) / 16

for i in range(5):
    img = cifar_x[i]

    resized = tf.image.resize(img, (8, 8)).numpy().astype(np.uint8)

    gray = np.mean(resized, axis=2)

    blur = tf.nn.conv2d(
        gray.reshape(1, 8, 8, 1).astype(np.float32),
        kernel,
        strides=1,
        padding="SAME"
    ).numpy().squeeze()

    gx = np.abs(np.diff(blur, axis=1))
    gy = np.abs(np.diff(blur, axis=0))

    gx = np.pad(gx, ((0, 0), (0, 1)))
    gy = np.pad(gy, ((0, 1), (0, 0)))

    edges = np.sqrt(gx**2 + gy**2)
    edges = (edges > 20) * 255

    resized_images.append(resized)
    gray_images.append(gray)
    blur_images.append(blur)
    edge_images.append(edges)

plt.figure(figsize=(15, 12))
```

```
for i in range(5):
    plt.subplot(5, 5, i + 1)
    plt.imshow(cifar_x[i])
    plt.title("Original")
    plt.axis("off")

for i in range(5):
    plt.subplot(5, 5, i + 6)
    plt.imshow(resized_images[i])
    plt.title("Resized")
    plt.axis("off")

for i in range(5):
    plt.subplot(5, 5, i + 11)
    plt.imshow(gray_images[i], cmap="gray")
    plt.title("Gray")
    plt.axis("off")

for i in range(5):
    plt.subplot(5, 5, i + 16)
    plt.imshow(blur_images[i], cmap="gray")
    plt.title("Blur")
    plt.axis("off")

for i in range(5):
    plt.subplot(5, 5, i + 21)
    plt.imshow(edge_images[i], cmap="gray")
    plt.title("Edges")
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Original Size: 1024 x 768

Resized (25%): 256 x 192



```
In [ ]:
```